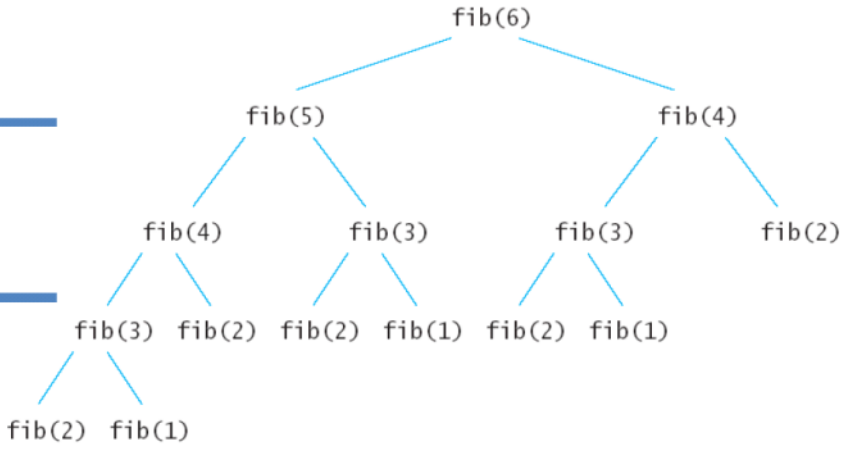
COMP 2121
DISCRETE MATHEMATICS

Lecture 12



Sequences

QUIZ 3 ON FSM

SET A OCT 27

SET B,C OCT 28

SET D OCT 30

Sequences defined by formula

Definition: *Sequence is an ordered list of numbers (or objects).*

- **Sequences defined explicitly**

Example 1. Given a formula, find the terms of $a_k = \dfrac{k}{k+1}, k \geq 1$ and $b_i = \dfrac{i-1}{i}, \quad i \geq 2$

$a_{\square}$

a_3 vs $a3$ vs a^3

$a_1 = \dfrac{1}{1+1} = \dfrac{1}{2}$

$a_2 = \dfrac{2}{2+1} = \dfrac{2}{3}$

$a_3 = \dfrac{3}{3+1} = \dfrac{3}{4}$

$b_2 = \dfrac{2-1}{2} = \dfrac{1}{2}$

$b_3 = \dfrac{3-1}{3} = \dfrac{2}{3}$

$b_4 = \dfrac{3}{4}$

Example 2. Given the first k terms, find the formula:

$\frac{1}{1}$, 1/4, 1/9, 1/16, 1/25, 1/36, …

$a_k = \dfrac{1}{k^2}$; $k \geq 1$

$a_i = \dfrac{1}{i^2}$; $i \geq 1$

$a_j = \dfrac{1}{(j-10)^2}$; $j \geq 11$

Alternating sequences

Example 3. List the terms of alternating sequence $\boxed{a_k = (-1)^k}$, $k \geq 0$ and find an explicit formula to describe the following sequence: $\underbrace{1,\ -1/4,\ 1/9,\ -1/16,\ 1/25,\ -1/36,}$ …

$a_0 = (-1)^0 = 1$

$a_1 = (-1)^1 = -1$

$a_2 = (-1)^2 = 1$

$$a_k = \frac{(-1)^{k-1}}{k^2} \quad ; \ k \geq 1$$

➢ Sequences defined recursively

Consider the following sequence of numbers:

$$\boxed{0, 1}, 1, 2, 3, 5, 8, 13, 21, 34 \ldots \quad 55$$

It is the case that each element of the sequence (other than the first two elements) is the sum of the previous two elements. For example, 2= 1+1; 21 = 8+13 and so on. This sequence is called *Fibonacci sequence*, and can recursively be defined as:

$a_n = a_{n-1} + a_{n-2}$, n>2 where a_1=0 and a_2=1.

```
int fib (int n) {
  if n=1
     return 0
  if n=2
     return 1
  if n>=3
  return fib(n-1) + fib(n-2)
}
```

$a_1 = 0$

$a_2 = 1$

$a_n = a_{n-1} + a_{n-2}; \; n \geq 3$

$a_5 = a_4 + a_3 = 2+1 = 3$

$a_4 = a_3 + a_2 = 1+1 = 2$

$a_3 = a_2 + a_1 = 1+0 = 1$

We have learned that $n! = n \cdot (n-1) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$. Factorial can be expressed recursively too. Let $a_n = n!$. Also, $a_{n-1} = (n-1)!$ and it follows that $a_n = n \cdot a_{n-1}$. However, we still need to define initial condition (i.e. the first element) and that is a_1=1.

$$7! = 7 \cdot \underbrace{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}_{6!}$$

```
int fact (int n) {
    k = 1
    for i=1 to n
        k = k * i
    return k
}
```

$a_1 = 1$

$a_n = n \cdot a_{n-1}; \; n \geq 2$

```
int fact (int n) {
  if n=1
     return 1
  if n>=2
     return n * fact(n-1)
}
```

Example 4

Example 4. Given that, $a_n=5a_{n-1}$, $n>1$ and $a_1=7$ find a_{100} .

$a_1 = 7$

$a_2 = 5(7)$

$a_3 = 5(5(7)) = 5^2(7)$

$a_4 = 5(5^2(7)) = 5^3(7)$

$\vdots$

$a_{100} = 5^{99}(7)$

Example 5

Example 5. Give a recursive definition for the sequence defined as $\boxed{a_n = 4n - 1, n \geq 1}$.

$$\boxed{\begin{aligned} &a_1 = 3 \\ &a_n = a_{n-1} + 4 \quad ; \ n \geq 2 \end{aligned}}$$

$a_1 = 4(1) - 1 = 3$

$a_n \sim a_{n-1}$

$4n = a_n + 1$

$a_2 = 4(2) - 1$

$a_{n-1} = 4(n-1) - 1$

$a_{n-1} = 4n - 4 - 1$

$a_{n-1} = a_n + 1 - 4 - 1$

$a_{n-1} = a_n - 4$

$a_n = a_{n-1} + 4$

Summations

➢ **Summations**

$$\sum_{i=1}^{5} i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$$

$$\sum_{i=3}^{5} (i+2) = (3+2) + (4+2) + (5+2)$$

We use the following notation:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \ldots + a_{n-1} + a_n$$

Note that k is a variable of summation or dummy variable. You could use a j instead:

$$\sum_{j=m}^{n} a_j = a_m + a_{m+1} + a_{m+2} + \ldots + a_{n-1} + a_n$$

We can sometimes simplify summation by ch~~anging the variable and~~ the limits of the range, for example:

$$\underbrace{\sum_{k=0}^{6}\frac{1}{k+1}} = \sum_{j=1}^{7}\frac{1}{j}$$

$$\sum_{k=0}^{6}\frac{1}{k+1} = \frac{1}{0+1} + \frac{1}{1+1} + \frac{1}{2+1} + \frac{1}{3+1} + \frac{1}{4+1} + \frac{1}{5+1} + \frac{1}{6+1}$$

$$= \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$$

$$\sum_{j=1}^{7}\frac{1}{j}$$

Change of variable

Example 6. Transform the following summation by making a change of variable: $j = i$ -1

$$\sum_{i=1}^{n-1} \frac{i}{(n-i)^2}$$

Product notation (PI)

- **<u>Multiplying terms of a sequence</u>**

Product Notation:

$$\prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdot \dots \cdot a_n$$

Example 8. Given that $a_k = \dfrac{k}{k+1}$, $k \geq 1$, find a closed formula $\prod_{k=3}^{n} a_k$.

$$(((((50+0)+1)+2)+3)\cdots)+9)$$

Example 7. Think of the variable of summation as the variable that controls a loop in a program

$$50+0+1+2+3+4+5+6+7+8+9$$

Version A:

50

```
n = 0;
for (i = 0; i < 10; i++)
  n = n + i;
```

$$50 + \sum_{i=0}^{9} i$$

Version B

50

```
n = 0;
for (j = 1; j <= 10; j++)
  n = n + (j - 1);
```

$$50 + \sum_{j=1}^{10} (j-1)$$

Version C

50

```
n = 0;
for (k = 17; k >7; k--)
  n = n + (17 - k);
```

$$50 + \sum_{k=8}^{17} (17-k)$$

Does each of the above code examples result in the same value of *n*? Which code example would be preferable?

- **Properties of Summations and Products**

If $a_m, a_{m+1}, a_{m+2}, \ldots$ and $b_m, b_{m+1}, b_{m+2}, \ldots$ are sequences of (real) numbers and c is any (real) number, then the following equations hold for any integer $n \geq m$

1.	$\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n} (a_k + b_k)$
2.	$c \cdot \sum_{k=m}^{n} a_k = \sum_{k=m}^{n} c \cdot a_k$
3.	$\cancel{\left(\prod_{k=m}^{n} a_k\right) \cdot \left(\prod_{k=m}^{n} b_k\right) = \prod_{k=m}^{n} a_k \cdot b_k}$

$$\sum_{i=1}^{3} (2a_i + b_i) = (2a_1 + b_1) + (2a_2 + b_2) + (2a_3 + b_3)$$

$$= 2a_1 + 2a_2 + 2a_3 + b_1 + b_2 + b_3$$

$$= \sum_{i=1}^{3} 2a_i + \sum_{i=1}^{3} b_i$$

$$2 \cdot (a_1 + a_2 + a_3)$$

$$2 \sum_{i=1}^{3} a_i$$

Example 9

Example 9. Given that $\boxed{\sum_{k=1}^{99} k^2} = 328,350$ calculate $\sum_{k=1}^{100} (2+k^2)$

$$\sum_{k=1}^{100} (2+k^2) = (2+1^2)+(2+2^2)+(2+3^2)+\dots+(2+99^2)+(2+100^2)$$

$$= \underbrace{\sum_{k=1}^{100} 2}_{2\times 100} + \sum_{k=1}^{100} k^2$$

$$= 200 + \sum_{k=1}^{100} k^2 = 200 + \boxed{\sum_{k=1}^{99} k^2} + 100^2$$

$$= 200 + 328,350 + 10,000$$

$$= 338,550$$

$$\sum_{i=a}^{b} \text{const} = (b-a+1) * \text{const}$$

no i

$$\sum_{i=1}^{5} i = 15 \qquad \sum_{i=1}^{5} j = 5j$$

Example 10

Example 10. Given the formula $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

$$\sum_{i=1}^{5} i = 1+2+3+4+5 = \frac{5(5+1)}{2} = 15$$

a) compute $25 + 26 + 27 + 28 + \ldots + 92 + 93$.

$$\sum_{i=25}^{93} i = \sum_{\cancel{i=1}}^{93} i - \sum_{i=1}^{24} i = \frac{93(94)}{2} - \frac{24(25)}{2}$$

$$= 4071$$

$$\sum_{i=1}^{n+5} i = \frac{(n+5)(n+5+1)}{2}$$

b) compute the sum $\sum_{i=3}^{2n+1}(4i-3)$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

const!

$$\sum_{i=3}^{2n+1}(4i-3) = \sum_{i=3}^{2n+1} 4i - \sum_{i=3}^{2n+1} 3$$

$$= 4\sum_{i=3}^{2n+1} i - ((2n+1)-3+1) * 3$$

$$= 4\left(\sum_{i=1}^{2n+1} i - \sum_{i=1}^{2} i\right) - (2n-1)(3)$$

$$= \cancel{4}^{2}\left(\frac{(2n+1)\overbrace{(2n+1+1)}^{2n+2}}{\cancel{2}} - \frac{2(2+1)}{\cancel{2}}\right) - (6n-3)$$

$$= 2(4n^2+4n+2n+2-6) - 6n+3$$

$$= 2(4n^2+6n-4) - 6n+3$$

$$= 8n^2 + 12n - 8 - 6n + 3$$

$$= 8n^2 + \underline{12n} - 8 \underline{- 6n} + 3$$

$$= 8n^2 + 6n - 5$$

Example 11. The following pseudocode sorts array A, consisting of n elements, in increasing order. How many time units does it take to execute the pseudocode?

****You should note that the two for loops are nested loops.*

```
(1)        for i = 0 to n-2 do {
(2)            min = i
(3)            for j = (i + 1) to n-1 do {
(4)                if (A[j] < A[min]) {
(5)                    min = j
(6)                }
(7)            }
(8)            swap A[i] and A[min]
(9)        }
```

time = 0
For i = 0 TO n-2 {
For j = i+1 TO n-1 {
time = time + 2
}
time = time + 12
}

$$\sum_{i=2}^{5} 3 = 3+3+3+3$$

Assume the following:

- ✓ It takes insignificant amount of time to execute any statement other than statements in lines (4) and (8);
- ✓ It takes 2 time units for a single execution of the statement in line (4): if (A[j] < A[min]);
- ✓ It takes 12 time units for a single execution of the statement in line (8): swap A[i] and A[min].

Use formula $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

cont.

$$\text{time} = 0 + \sum_{i=0}^{n-2}\left[\sum_{j=i+1}^{n-1}[2] + 12\right]$$

$$= \sum_{i=0}^{n-2}\left[\left[(n-\cancel{1})-(i+1)+\cancel{1}\right]2 + 12\right]$$

$$= \sum_{i=0}^{n-2}\left[(n-i-1)(2)+12\right]$$

$$= \sum_{i=0}^{n-2}\left[2n-2i-2+12\right] = \sum_{i=0}^{n-2}\left[\underbrace{2n+10}_{\text{const}}-2i\right]$$

$$= \sum_{i=0}^{n-2}(2n+10) - \sum_{i=0}^{n-2} 2i \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$= (n-2+0+1)(2n+10) - 2\sum_{i=0}^{n-2} i$$

$$- \left(0 + \sum^{n-2} i\right)$$

$$= (n-1)(2n+10) - 2\left(0 + \sum_{i=1}^{n-2} i\right)$$

$$= 2n^2 + 10n - 2n - 10 - \cancel{2} \cdot \frac{(n-2)(n-2+1)}{\cancel{2}}$$

$$= 2n^2 + 8n - 10 - (n-2)(n-1)$$

$$= 2n^2 + 8n - 10 - (n^2 - n - 2n + 2)$$

$$= 2n^2 + 8n - 10 - (n^2 - 3n + 2)$$

$$= 2n^2 + 8n - 10 - n^2 + 3n - 2$$

$$= n^2 + 11n - 12$$